

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (EE)

Nov 2013

EE-204/EE-204.01 Electrical Power System-I

Date: 13.11.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain Single line diagram of A.C. power supply scheme with neat sketch. [07]
- Q - 2 (a) Define String efficiency. Explain any one method of improving it. [06]
- (b) What do you mean by Safety factor? Explain the causes of insulator failure in Pin-type insulator with necessary diagram. [05]
- (c) Give the comparison between Aluminum Conductor and ACSR conductor. [03]

OR

- Q - 2 (a) Give the classification of cables according to the voltage level for which they are manufactured. Explain construction of 3-Core cable with neat diagram. [07]
- (b) Give the desirable properties of Insulating material. Explain different insulating material used for cables. [07]
- Q - 3 (a) What do you mean by grading of cables? Explain Capacitance grading with neat sketch. [07]
- (b) The equipment in a power station costs Rs 15, 60,000 and has a salvage value of Rs 60,000 at the end of 25 years. Determine the depreciated value of the equipment at the end of 20 years on the following methods : [07]
- (i) Straight line method ; (ii) Diminishing value method ; (iii) Sinking fund method at 5% compound interest annually.

OR

- Q - 3 (a) What is Tariff? Give the desirable properties of tariff. [05]

- (b) Give the name of different techniques for Power factor improvement and explain any one in detail. [05]
- (c) Define the following terms. [04]
- (1) Demand factor
 - (2) Diversity factor
 - (3) Load factor
 - (4) Load curve

SECTION - II

- Q - 4 (a) Derive equation of Loop Inductance of a 1- Φ , 2-Wire line. [07]
- Q - 5 (a) Draw and explain Nominal π method of Medium transmission line with its vector diagram. [07]
- (b) Determine Generalized circuit constants (A, B, C, D) for Nominal T method. [07]

OR

- Q - 5 (a) A (medium) single phase transmission line 100 km long has the following constants : [07]
- Resistance/km = 0.25Ω ; Reactance/km = 0.8Ω
- Susceptance/km 14×10^{-6} siemen ; Receiving end line voltage = 66,000 V
- Assuming that the total capacitance of the line is localized at the receiving end alone, determine (i) the sending end current (ii) the sending end voltage (iii) regulation and (iv) supply power factor. The line is delivering 15,000 kW at 0.8 power factor lagging.
- (b) Give the disadvantages of low power factor. [05]
- (c) Explain Self G.M.D. and Mutual G.M.D. [02]
- Q - 6 (a) Derive the equation of Inductance of a 3-phase overhead line with unsymmetrical spacing. [07]
- (b) Write short note on Bundle conductor. [05]
- (c) Give the importance of Transposition of transmission line. [02]

OR

- Q - 6 (a) Explain the End condenser method with vector diagram and give the limitation of this method. [07]
- (b) Find the inductance per phase per km of double circuit 3-phase line having: [07]
Horizontal distance between 2 adjacent conductors is 6 mm.
Vertical distance between 2 adjacent conductors is 3 mm.
The conductors are transposed and are of radius 0.75 cm each. The phase sequence is ABC.
